30 Years After 51 Pegasi b

How Exoplanets Changed the way We Viewed the Universe

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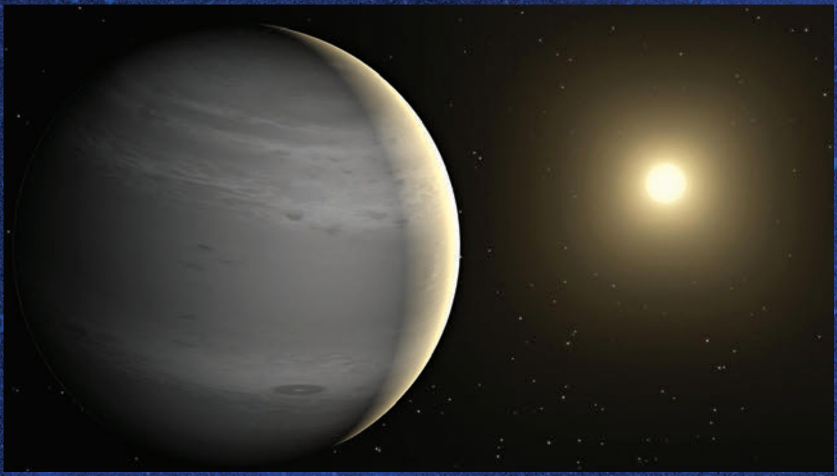

Thirty years ago, on a crisp October day in 1995, two Swiss astronomers changed the course of cosmic history. Michel Mayor and Didier Queloz, working at the University of Geneva, announced the discovery of 51 Pegasi b – the first planet found orbiting a Sun-like star beyond our solar system. It was a startling revelation: not only did it confirm that other worlds existed, but it also shattered long-held assumptions about how planets formed.

51 Pegasi b was no Earth-like paradise. It was a “hot Jupiter” – a massive gas giant orbiting so close to its star that a year lasted just 4.2 days. To astronomers of the 1990s, this made no sense. According to classical models, gas giants should form far from their stars, like Jupiter and Saturn in our own solar system. Yet here was a planet that defied the rules, forcing scientists to rethink everything they thought they knew about planetary systems.

From wobbles to worlds

The first detection relied on a subtle “stellar wobble.” As a planet orbits its star, its gravity tugs the star ever so slightly, shifting the star’s light spectrum back and forth. Mayor and Queloz used the ELODIE spectrograph to measure these minuscule variations in starlight, just a few meters per second, revealing the invisible presence of a planet.

Since then, the technology has advanced at an astonishing pace. Instruments like HARPS and ESPRESSO, built by European teams including NCCR PlanetS in Switzerland, can now measure stellar motions with a precision of just 10 centimeters per second. At the same time, space telescopes like Kepler and TESS have revolutionized planet hunting through the transit method, detecting the tiny dip in brightness that occurs when a planet crosses in front of its star.

A cosmic mirror

The search for another Earth has become both a scientific and philosophical journey. Astronomers now estimate that there may be billions of Earth-sized planets in our galaxy alone. Some, like Kepler-452 b and Proxima Centauri b, orbit within their stars’ habitable zones and could, in theory, support liquid water. Yet the challenge lies in confirming whether these worlds truly resemble our own or if they’re merely rocky husks scorched or frozen by their stars.

Future missions such as PLATO, LUVOIR, and Ariel aim to peer deeper into exoplanet atmospheres, searching for signs of life: oxygen, methane, water vapor. With instruments detecting these faint chemical fingerprints, humanity may just answer the most profound question of them all – are we alone? d